



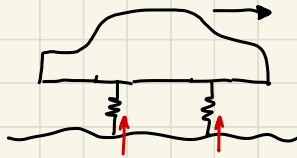
ESE 4 particular kind of excitation

which may occur in many cases of mech systems

→ When instead of excitation act as $f(t)$ shaking the system
We HAVE A MOTION applied on the constraints of the syst.

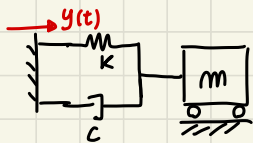
Constrain movement. ⇒ for EXAMPLE road vehicle

Simplest case
of constraint motion



excitation associated
to the wheel dry
motion

es 1



instead of having $f(t)$ act on system, we
assume constrain clamp on wall)
moving according to $y(t)$ KNOWN
(deterministic "noise")

x Indip. coordinate of slider (absolute position)

With LAGRANGE

$$\begin{cases} T = \frac{1}{2} m \dot{x}^2 \\ V = \frac{1}{2} K (x - y)^2 \\ D = \frac{1}{2} c (\dot{x} - \dot{y})^2 \end{cases} \Rightarrow \underbrace{\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right)}_{m \ddot{x}} - \underbrace{\frac{\partial T}{\partial x}}_0 + \frac{\partial D}{\partial \dot{x}} + \frac{\partial V}{\partial x} = \underbrace{\frac{\delta W}{\delta x}}_{f^*(t)}$$

$$m \ddot{x} + c(\dot{x} - \dot{y}) + K(x - y) = Q$$

NO Time dependent force → $\delta W = Q$
We will correct this

$$m \ddot{x} + c \dot{x} + Kx = c \dot{y} + Ky$$

↑ unknown $x(t)$ KNOWN function of time $y(t), \dot{y}(t)$

due to moving constrain (like in an accelom.) ← { equivalent time dependent forcing $f^*(t)$ }

so...

a motion applied in

Some point of the

System is equivalent to a force applied on the system representing

an acting excitation equivalently → deal with excitation coming from

forces OR from system motion applied

(es1) is a simple case that can be easily solved (also with newton II law)

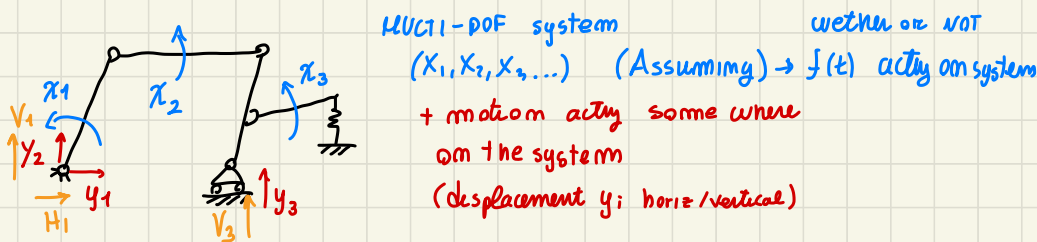
IF more **COMPLEX** system

↓
We need a GENERAL APPROACH, obtaining a VECTOR NOTATION
ending up to a general system approach $\forall$ N -DOF system
- finite number DOF

subjected to any "m" displacement components

Specify this EXTENSION approach, for writing eq of motion linear

In a complex general case



We have general procedure TO deal WITH IT! $\Rightarrow$ We need to add something

EXTENDED
VECTOR of
INDIP. coord
(general case)

$$\underline{x} = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \\ \hline y_1 \\ y_2 \\ \vdots \\ y_m \end{Bmatrix}$$

independent coord for syst motion $\swarrow$ unhegging of process
to describe point displacement excitation MOTION

Setting the

motion $y_1 = y_2 = \dots = y_m = 0$

classical motion. If instead we have displacement applied effect $\rightarrow$ we develop the linearized eq of MOTION according to this extended set of coordinates (including displacement

if dividing the $\underline{x}$ coord + real coord dim.)

between $\underline{x}, \underline{y} \Rightarrow (\underline{x}_1 \dots \underline{x}_m)^T$ is UNKNOWN scalar variables!

$(\underline{y}_1 \dots \underline{y}_m)^T$ KNOWN displacement applied to the system

(we assume we know a priori the forced displacem. time history of motion known)

$\underline{x}$ mix
known and
unknown parts...

$$\underline{x} = \begin{Bmatrix} \underline{x}_f \\ \underline{x}_c \end{Bmatrix}$$

Free to obtain any value
constrained by a specific motion applied y_i

With lumping procedure

$$\underline{x} = \begin{Bmatrix} \underline{x}_f \\ \underline{x}_c \end{Bmatrix}_{m+m} \quad \begin{array}{l} \text{vector of } m+m \text{ indep coord} \\ \downarrow \\ \text{we end up with} \end{array}$$

done with same
general theory
approach!

$$[m] \ddot{\underline{x}} + [c] \dot{\underline{x}} + [k] \underline{x} = \underline{f}(t)$$

$m+m$ DOF
diff eq

↑ we get an equation NOT
ready to be solved, BECAUSE

- on $\underline{x}$ we have a MIX of UNKNOWNs / KNOWNs part...
We have to provide a vector $\underline{x}$ with only UNKNOWNs
- and also $\underline{f}$ is a MIX of two things → effect of time-dependent forces
acting directly to system + (because of $\underline{x}$ extension with forced displacement)
so the constraint forces $N, V...$ causes a NON ZERO δW
 V_1, H_1, V_3 can produce WORK so inside $\underline{f}(t)$ we have
(also virtual work due to δy_i ... last ROWs as
unknown terms! ISSUE)

MIX of KNOWN / UNKN both on $\underline{x}$ and $\underline{f}$
to solve this, as we can partition $\underline{x}$
We do the same for other terms

$$\underline{x} = \begin{pmatrix} \underline{x}_f \\ \underline{x}_c \end{pmatrix} \begin{array}{l} \uparrow m \\ \downarrow m \end{array} \quad (\text{split vector})$$

$$[m] = \begin{bmatrix} [m_{ff}] & [m_{fc}] \\ [m_{cf}] & [m_{cc}] \end{bmatrix} \begin{array}{l} m, m \\ m, m \\ m+m, m+m \end{array}$$

split $[m]$ into 4 BLOCKS
submatrix
separation by: separating first m ROWs
and m column

We do the same with $[c]$, $[k]$ and
also SPLIT

$$\underline{f}(t) = \begin{Bmatrix} \underline{f}_f(t) \\ \underline{f}_c(t) \end{Bmatrix} \begin{array}{l} \uparrow m \\ \downarrow m \end{array}$$

↑ Reaction forces!

Look again to our matrix equation of syst dynamic,
 once split $\underline{x}, \dot{\underline{x}}, \ddot{\underline{x}}$ and $[m], [c], [k], \underline{f}$

WE SPLIT the equation of motion:

into 2 equations (and dividing into BLOCKS $[m], \dots$ ecc)

we can use each block as a single "scalar terms" providing consistency between sizes

• (KNOWN A PRIORI)

$$\begin{cases} [m_{ff}] \ddot{\underline{x}}_f + [m_{fc}] \ddot{\underline{x}}_c + [c_{ff}] \dot{\underline{x}}_f + [c_{fc}] \dot{\underline{x}}_c + [k_{ff}] \underline{x}_f + [k_{fc}] \underline{x}_c = \underline{f}_f & (1) \\ [m_{cf}] \ddot{\underline{x}}_f + [m_{cc}] \ddot{\underline{x}}_c + [c_{cf}] \dot{\underline{x}}_f + [c_{cc}] \dot{\underline{x}}_c + [k_{cf}] \underline{x}_f + [k_{cc}] \underline{x}_c = \underline{f}_c & (2) \end{cases}$$

take first m rows in the larger eq
 east m rows of smaller equation

using VECTOR-MATRIX notation (looking eq. (1))

we notice KNOWN terms $\underline{x}_c \rightarrow \dot{\underline{x}}_c \rightarrow \ddot{\underline{x}}_c$ a priori known displacement

↓ we move it in the right Forces directly applied (m scalar eq.)

$$[m_{ff}] \ddot{\underline{x}}_f + [c_{ff}] \dot{\underline{x}}_f + [k_{ff}] \underline{x}_f = \underline{f}_f - ([m_{fc}] \ddot{\underline{x}}_c + [c_{fc}] \dot{\underline{x}}_c + [k_{fc}] \underline{x}_c)$$

↑ equation we know how to solve (free/excited motion)

related to free motion

linear MATRIX eq. (superimposition)

↑ effect of motion applied to the system!

we super impose

the effect of time dep. forces and displacement of constraints (sum of the two effects)

Looking equation (2) → what we can do with it?

(m scalar equation) as amount of displacem. forced...

We can use (2) to obtain the unknowns related to REACTION FORCES

once solved (1) we have $\underline{x}_f$, so we plug-in

$$\underline{x}_f, \dot{\underline{x}}_f, \ddot{\underline{x}}_f + \underline{x}_c, \dot{\underline{x}}_c, \ddot{\underline{x}}_c \rightarrow \text{we can find } \underline{f}_c(t) \text{ and}$$

(solved) (yet known)

extract REACTION forces from it

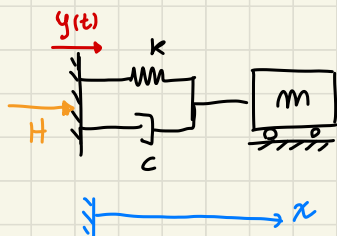
(When we need the REACTION forces! maybe as

control eng. we could need to design the actuator producing motion, or shaking machine creating motion vibration)

(for example done on funny activity on manufacturing)

you may design the force of displacement (Reaction)

We can exemplify the process in a simple 2DOF oscillator



$y(t)$ applied displacement of wall

x as indep coord to describe absolute slider motion

simple 2DOF eq of motion with $\underline{x} = \begin{pmatrix} x \\ y \end{pmatrix}$
 extended vector of indep coord with x and single displacem. motion applied y

$$\begin{bmatrix} m & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix} + \begin{bmatrix} c & -c \\ -c & c \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} + \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ H \end{bmatrix}$$

horiz reaction force H producing VIRTUAL WORK

$\underline{x} = \begin{bmatrix} x \\ y \end{bmatrix} \leftarrow \begin{matrix} x_f \\ x_c \end{matrix}$ split vector, easy partition also for MATRIX

↓ (specifying the equation of the CONSTRAINT / indep coh.)

⇒ We specify (1st eq) as

$$m\ddot{x} + c\dot{x} + kx = c\dot{y} + ky$$

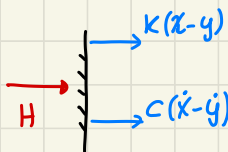
exactly as previously (1)

While from (2nd eq) in this case:

$$-c\dot{x} + c\dot{y} - kx + ky = \bar{H} \quad \text{Unknown} \Rightarrow H = -c(\dot{x} - \dot{y}) - k(x - y) \quad (*)$$

easy to see from constrain forces

H should balance what act on the wall



(H < 0)

NO mass associated D'ALAMB.

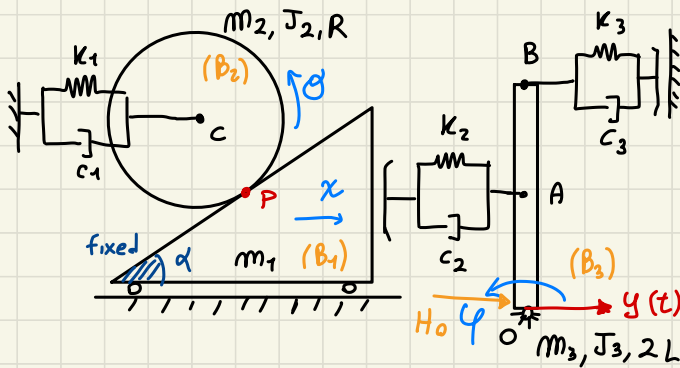
$$\sum F_x = 0 \quad (*)$$

this described

procedure works well ...

luckily this work & general case difficult to analyze with scalar approach... easy generalization & case

es2 (application of what we see)



- made up by
- 3 R.B 9 DOF
 - slider B1 -2 DOF
 - Rolling with sliding of B2 -2 DOF
 - Hinge on B3 -2 DOF
- ↓
- 3 DOF**

- We prefer as indep coord : x, θ, φ
- With excitation as displacement y applied in the pendulum hinge
 $\Rightarrow$ moves of $y(t)$ HORIZONTALLY (JUST function of time $y(t)$)

WE CARE about Writing eq. of motion for this system:

With VECTOR NOTATION (a priori $m=3$, $m=1$ additional extended vector)

$$\left\{ \begin{array}{l} \text{extended} \\ \text{vector} \end{array} \right\} \quad \underline{x} = \begin{Bmatrix} x \\ \theta \\ \varphi \\ y \end{Bmatrix} = \begin{Bmatrix} \underline{x}_f \\ \underline{x}_c \end{Bmatrix} \quad \begin{array}{l} m=3 \\ m=1 \end{array}$$

LAGRANGE

in matrix form... (try to evaluate energetic quantities)

$$T = \frac{1}{2} \dot{\underline{y}}_m^T [\text{mph}] \underline{y}_m \quad \text{here} \quad \dot{\underline{y}}_m = \begin{Bmatrix} v_{G1,x} \\ v_{G2,x} \\ v_{G2,y} \\ \omega_2 \\ v_{G3,x} \\ \omega_3 \end{Bmatrix}$$

velocity component needed for T

consistently

$$[\text{mph}] = \text{diag} \{ m_1; m_2; m_2; J_2; m_3; J_3 \}$$

find Jacobian matrix

$$\dot{\underline{y}}_m = [\lambda_m] \dot{\underline{x}}$$

We don't care if m
 $\&$ we have real coh. or displacement $y \in x_c$

We will consider this later!

- B1 moves only along x
- B2 ROTATE + moving along x, y (L)
- B3 moves along x (A)
but $v_{G3,y} = 0$ because encastré
here ROTATE

λ_{lm} with Rows as $\dot{y}_m$ and 4 columns as $\dot{x}$

$$[\lambda_{lm}] = \begin{bmatrix} \dot{x} & \dot{\theta} & \dot{\varphi} & \dot{y} \\ +1 & 0 & 0 & 0 \\ 0 & 1-R\cos(\alpha) & 0 & 0 \\ 0 & -R\sin(\alpha) & 0 & 0 \\ 0 & 0 & +1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} V_{G1x} \\ V_{G2x} \\ V_{G2y} \\ \omega_2 \\ V_{G3x} \\ \omega_3 \end{matrix}$$

Fill it with
super-Imposition
principle, on
a different cases...



• $\dot{x} \neq 0; \dot{\theta}, \dot{\varphi}, \dot{y} = 0$

slide triangle moves as TRANSLATION $\dot{x}$

θ frozen, so the disk TRANSLATE with same $\dot{x}$ velocity as point P

$v_P = \dot{x}$ (NO Relative sliding between this)

so same $v_C = \dot{x}$ cause $\dot{\theta} = 0$

While along y no disk motion θ is frozen. so the bar is still

if cannot ROTATE

• $\dot{\theta} \neq 0, \dot{x} = \dot{\varphi} = \dot{y} = 0$

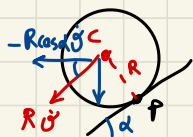
Bar fixed... $V_{G3x} = \omega_3 = 0$

triangle body still (frozen)

While P as disk point $\rightarrow$ same

of P triangle $v_P = 0 \rightarrow$ P is the C.I.R

of the disk rotation of $\dot{\theta}$



direction with
right hand
principle...

• $\dot{\varphi} \neq 0, \dot{x} = \dot{\theta} = \dot{y} = 0$

still triangle, $\dot{x} = 0$

$\dot{\theta} = 0 \rightarrow$ disk can just TRANSLATE if triangle moves, BUT it is frozen (Disk frozen!)

because $\dot{y} = 0$ point O still $\rightarrow O \equiv$ C.I.R of BAR

so $v_A = L \dot{\varphi}$



• $\dot{y} \neq 0, \dot{x} = \dot{\theta} = \dot{\varphi} = 0$

still Disk and triangle still

and $\omega_3 = 0$, NOT ROTATE

BAR JUST moving translates

once expressed

$[\lambda_{lm}] \dots$

taking all the

product $[\lambda_{lm}]^T [M_{lph}] [\lambda_{lm}] = [c_m]$

$\rightarrow$

expressing the total mass matrix 4×4 [m] to see the PARTITIONING needed

$$[m] = \begin{bmatrix} m_1 + m_2 & -m_1 R \cos \alpha & 0 & 0 \\ -m_1 R \cos \alpha & m_2 R^2 + J_2 & 0 & 0 \\ 0 & 0 & m_3 L^2 + J_3 & -m_3 L \\ 0 & 0 & -m_3 L & m_3 \end{bmatrix}$$

[m_{ff}]
[m_{fc}]
[m_{cf}]
[m_{cc}]

separating the rows / column

we have all needed to WRITE inertia for the two equations of free cohort and reaction forces act on D!

POTENTIAL ENERGY:

$$V = V_{ee} + V_g$$

as usual comes from elastic terms (k_1, k_2, k_3) and weight force gravitation

for sure $[K_I]$ is present ..

there is any $[K_{II}]$? II ORDER elastic stiffness

← NO

Don't care about $[K_{II}]$

because the relationship between the Δe and

indip cohort is LINEAR OR it depend non linearly with $\sin(\text{cohort})$ like k_3 as $-2L \sin(\varphi) \rightarrow \frac{\partial^2}{\partial \varphi^2} \rightarrow 0$ evaluated @ eq.

instead what about $[K_g] \Rightarrow$ NO vertical movement of triangle

neither the disk get contriutes cause y_c linearly depends on θ

(wheel came out) $\rightarrow$ BUT point A of BAR depends non linearly by φ with $\cos(\varphi) \rightarrow$ NON ZERO GRVIT.

STIFFNESS due to THE BAR!

$$V_{el} = \frac{1}{2} \underline{y}_k^T [K_{ph}] \underline{y}_k \quad \underline{y}_k = \begin{Bmatrix} \Delta e_1 \\ \Delta e_2 \\ \Delta e_3 \end{Bmatrix}$$

$$[K_{ph}] = \text{diag} \{ k_1, k_2, k_3 \}$$

with sign convention

← mm → (+)

to express $[\lambda_k] \Rightarrow$

$$[\lambda_k] = \begin{bmatrix} x & \theta & \varphi & y \\ +1 & -R \cos \alpha & 0 & 0 \\ -1 & 0 & -L & 1 \\ 0 & 0 & 2L & -1 \end{bmatrix} \begin{matrix} \Delta \ell_1 \\ \Delta \ell_2 \\ \Delta \ell_3 \end{matrix} \quad (\text{Super-imposition approach})$$

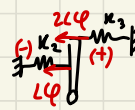
• $x \neq 0, \theta = \varphi = y = 0$

disk moves as triangle $\Delta \ell_3 = 0$ cause depends only on $\bar{P}$ on

• $x = 0, \theta \neq 0, \varphi = y = 0$

$\bar{P}$ is fixed, also triangle frozen, but $\Delta \ell_1 \neq 0$ cause change of length due to disk ROTATION

• $x = \theta = 0, \varphi \neq 0, y = 0$

triangle + disk still while BAR rotate  (small motion linearize!)

• $x = \theta = \varphi = 0, y \neq 0$

BAR moves horizontally, so k_2, k_3 extend/compress

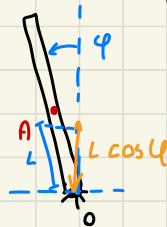
So we can evaluate

$$[K_{el}] = [\lambda_k]^T [K_{ph}] [\lambda_k] = \begin{bmatrix} k_1 + k_2 & -k_1 R \cos \alpha & k_2 L & -k_2 \\ -k_1 R \cos \alpha & k_1 R^2 \cos^2 \alpha & 0 & 0 \\ k_2 L & 0 & k_2 L^2 + 4k_3 L^2 & -k_2 L - 2k_3 L \\ -k_2 & 0 & -k_2 L - 2k_3 L & k_2 + k_3 \end{bmatrix} \begin{matrix} [K_{fc}] \\ [K_{fc}] \\ [K_{fc}] \\ [K_{fc}] \end{matrix}$$

partition of this $[K_{el}]$
 We should consider also the GRAVITATIONAL $[K_g]$ to be descent

$$V_g = m_3 g \underbrace{h_3}_{\substack{\text{height of BAR} \\ \text{center of mass} \\ \text{A given} \\ \text{ref level @ 0}}}$$

JUST for PENDULUM, the others is const or linearly vary!



$$V_g = m_3 g L \cos \varphi \approx m_3 g L \left(1 - \frac{\varphi^2}{2}\right)$$

only effect of φ
 involving y changes!
 to get linearized effect with power series expansion @ 1st ORD $\cos \varphi \approx \left(1 - \frac{\varphi^2}{2}\right)$

$$V_g = m_3 g L - \frac{1}{2} m_3 g L \varphi^2$$

$\frac{\partial}{\partial \varphi}$ deriving it we obtain from constant terms of K_g

$$V_g \approx -\frac{1}{2} m_3 g L \varphi^2 \quad (\text{pure quadratic term in } \varphi^2)$$

$$\approx \frac{1}{2} \underline{x}^T [K_g] \underline{x}$$

$$[K_g] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -m_3 g L & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

so the overall

$[K_{TOT}]$ is JUST

$[K_{el}] + [K_g]$ we have only effect on $[K_{33}]$

↓

partitioning

$$[K_{TOT}] = \begin{bmatrix} K_1 + K_2 & -K_1 R \cos \alpha & K_2 L & -K_2 \\ -K_1 R \cos \alpha & K_1 R^2 \cos^2 \alpha & 0 & 0 \\ K_2 L & 0 & K_2 L^2 + 4K_3 L^2 - m_3 g L & -K_2 L - 2K_3 L \\ -K_2 & 0 & -K_2 L - 2K_3 L & K_2 + K_3 \end{bmatrix}$$

$[K_{ff}]$ $[K_{fc}]$
 $[K_{cf}]$ $[K_{cc}]$

Respect DAMPING here is easy to see how dampers are all parallel to spring so

$$[C] = [K_{el}] \quad \text{changing } \begin{Bmatrix} k_1 \\ k_2 \\ k_3 \end{Bmatrix} \rightarrow \begin{Bmatrix} c_1 \\ c_2 \\ c_3 \end{Bmatrix}$$

you get some c matrix to partition as K_{el}

$$[C] = \begin{bmatrix} c_1 + c_2 & -c_1 R \cos \alpha & c_2 L & -c_2 \\ -c_1 R \cos \alpha & c_1 R^2 \cos^2 \alpha & 0 & 0 \\ c_2 L & 0 & c_2 L^2 + 4c_3 L^2 & -c_2 L - 2c_3 L \\ -c_2 & 0 & -c_2 L - 2c_3 L & c_2 + c_3 \end{bmatrix}$$

$[C_{ff}]$ $[C_{fc}]$
 $[C_{cf}]$ $[C_{cc}]$

virtual work: we have NO $f(t)$ time dep force
but only excitation due to $y_c(t)$ hinge displacement

$$\delta W = H_0 \delta y = \delta \underline{x}^T \underline{f}(t)$$

↑
no time
dep force effect → we have the H_0 Reaction force
acting on 0 compared to $\delta W = 0$
with $y \neq 0$ cohort

$$\underline{f}(t) = \left\{ \begin{array}{l} 0 \\ 0 \\ 0 \end{array} \right\} \left\{ \begin{array}{l} \delta x \\ \delta \theta \\ \delta \varphi \end{array} \right\} \left\{ \underline{f}_s \right\} \text{ (partition)}$$

$$\left\{ \begin{array}{l} H_0 \\ \delta y \end{array} \right\} \left\{ \underline{f}_c \right\}$$

all ingredients to
write the two equations
related to indep cohort and constraint forces

1st eq to provide
the movement x, θ, φ
while 2nd eq provides the Reaction force $H_0(t)$
(useful on DESIGN the actuation system)

↓
solve this equation by MATLAB (computer)